# Danny Muir

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## EDUCATION

**University of Washington, Seattle** *Expected June 2027*
Bachelor of Science in Computer Engineering — **GPA: 3.84** Dean's List All Quarters 2023–2026
Paul G. Allen School of Computer Science & Engineering | **Minor:** Neural Computation and Engineering
**Honors:** Tau Beta Pi Engineering Honor Society; CSE Honors

## TECHNICAL SKILLS

**Programming:** Python, Java, C, JavaScript, Bash, Git, HTML, LaTeX | **Platforms:** macOS, Linux, Windows
**ML/AI:** PyTorch, NumPy, Pandas, scikit-learn, MAML/Meta-Learning, Test-Time Adaptation, Autoencoders, Neural Networks, Computer Vision, Transformers
**Neuroscience:** ECoG/cortical probe signal processing, sEMG, neural decoding, temporal basis function models, closed-loop BCIs
**Research:** Grant writing, experimental design, project management, scientific computing, technical documentation

## PUBLICATIONS

M. J. Bryan, **D. C. Muir**, F. Schwock, A. Yazdan-Shahmorad, R. P. N. Rao, "Robust Neural Stimulation Response Modeling Through Meta-Learning and Pretraining," *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (Submitted; arXiv:2608.26649).

## RESEARCH EXPERIENCE

**Undergraduate Researcher** Oct 2025–Present
*Neural Systems Laboratory, UW*
*Advisor: Dr. Rajesh Rao, Paul G. Allen School of Computer Science & Engineering*

- Optimizing multi-session neural decoding framework in collaboration with Matthew Bryan, building on temporal basis function models (TBFMs) for closed-loop neural stimulation.
- Testing and selecting optimal implementations for session-specific embeddings and test-time adaptation methods.
- Improving training efficiency, model complexity, and cross-session generalization through systematic empirical evaluation.
- Conducting honors thesis research on learning dynamical systems underlying traveling waves in motor cortex to forecast motor-relevant neural subspaces.
- Extending Active Predictive Coding framework with hypernetwork-based hierarchical architectures to model cognitive map formation and entorhinal-hippocampal interaction.

**Associate Researcher, VP Operations** Jan 2025–Oct 2025
*Sensors, Energy, and Automation Laboratory (SEAL), UW*
*Advisor: Dr. Alexander Mamishev, Department of Electrical & Computer Engineering*

- Conducted VLF/ULF signal processing research for atmospheric and geophysical monitoring applications.
- Contributed written sections and managed submission logistics for federal and industry grant proposals (DOE STTR, NIH, NSF, CDC, Google).
- Led project management team developing custom automation and DevOps solutions for laboratory operations.

## GRANTS AND FUNDING

**Federal Research Proposals (2025):** Contributed written sections and managed submission logistics for DOE STTR, NIH, NSF, and CDC proposals on automation, sensor technology, and biomedical applications

**Industry Research Proposals (2025):** Contributing researcher on Google and industry partner applications for ML and automation research

## LEADERSHIP AND SERVICE

**President** Nov 2025–Present

*Triangle Fraternity, Washington Chapter*

- Established Triangle UW's first Interfraternity Council (IFC) membership, expanding chapter participation in campus Greek life programming, philanthropy infrastructure, and inter-chapter relationships.
- Scaled sorority collaborations from ∼1 event per quarter to 11 events across two quarters (4 partner chapters, 2 new); chapter grew from 21 to 40 members with near-complete retention.
- Rebuilt executive board from 6 to 8+ formalized roles — adding Housing and Communications Directors and formalizing Academic Chair and Rush Chairs — distributing operational ownership and building institutional continuity.
- Chapter GPA above all-IFC and all-men's benchmarks every quarter; 66% of members on Dean's List; 100% new member attendance across all IFC conduct programs; 125+ combined service hours for the Leukemia & Lymphoma Society.

**Vice President & New Member Educator** Dec 2023–Nov 2025

*Triangle Fraternity, Washington Chapter*

- Built and systematized chapter event operations, enabling consistent execution of social and brotherhood programming.
- Served as chapter academic lead; chapter earned the Kahlert Academic Excellence Award (highest GPA of any Triangle chapter nationally) and Most Improved Chapter recognition from national Triangle.
- Redesigned new member education program, improving integration processes and developmental outcomes.